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Page Rank

What is PageRank?

PageRank is a link analysis algorithm that measures the relative importance of web pages within a network (like the Internet). It was developed by Larry Page and Sergey Brin in 1998 as part of the original Google search engine.

The core idea is:

“A page is important if many important pages link to it.”

Intuition behind PageRank

Imagine a random surfer who:

- Starts on any web page.
- Randomly follows one of the links on that page.
- With some probability (e.g., 15 %), gets bored and jumps to any random page on the web.

After many steps, the frequency with which the surfer visits each page reflects its importance.

Mathematical formula

The classic PageRank formula for a page P_i is:

$$PR(P_i) = \frac{1-d}{N} + d \sum_{P_j \in M(P_i)} \frac{PR(P_j)}{L(P_j)}$$

Where:

- $PR(P_i)$: the PageRank of page i .
- d : the **damping factor** (usually 0.85).

- N : total number of pages.
- $M(P_i)$: the set of pages that link to P_i .
- $L(P_j)$: number of outbound links on page j .

What does this formula mean?

$$\frac{1-d}{N} : \text{probability of jumping to a random page.}$$

Mathematical Interpretation

PageRank can be formulated as an **eigenvector problem** of a **stochastic matrix**:

$$\vec{r} = dA\vec{r} + \frac{1-d}{N}\vec{1}$$

- $\vec{r}$: the PageRank vector.
- A : transition matrix (derived from the link graph).
- $\vec{1}$: vector of ones.

This equation can be solved via **power iteration**.

How is it Computed?

It uses an iterative process:

1. Initialize all PageRanks to $\frac{1}{N}$.
2. Apply the formula repeatedly until the values stabilize (converge).

This method scales well for millions of pages.

Simple Example

Imagine 3 pages:

- A links to B
- B links to C
- C links to A

With damping factor $d = 0,85$, the PageRanks converge to equal values, since the flow between pages is cyclic and balanced.

Important Properties

- **Global:** measures relative importance among all pages.
 - **Recursive:** a page's value depends on the pages linking to it.
 - **Spam-resistant:** hard to artificially inflate with irrelevant links.
 - **Interpretable:** easy to understand as a probability of visiting.
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Modern Applications

Although Google now uses more sophisticated methods, the idea of PageRank is still widely used in:

- **Network science** (e.g., social, biological, transportation networks)
 - **CiteRank** (ranking scientific publications)
 - **Recommendation systems**
 - **Spam and fraud detection**
 - **Social media influence analysis**
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Algorithm Variants

- **Personalized PageRank:** random jumps favor certain pages (e.g., personalized recommendations).
 - **Topic-sensitive PageRank:** multiple PageRank vectors for different topics.
 - **TrustRank:** incorporates trusted sources to combat low-quality content.
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Summary

PageRank is an elegant and powerful algorithm that uses link structure to estimate the relative importance of web pages or other nodes in a network. It was the foundation of Google's early success and still inspires modern graph analysis techniques.
